THE CONORMAL SEQUENCE OF TWO FORMALLY UNRAMIFIED MORPHISMS WITH A FORMALLY SMOOTH COMPOSITE IS SHORT EXACT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5015-5036 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5038-5076, the complete author proof (37 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings, conormal sequences of a pair of immersions, and formally smooth morphisms; the comparison is retained.

Source reference. Original lemma lemma-two-unramified-morphisms: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3392–3412. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universally-unramified-differentials-sequence: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3366–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ & \searrow^{j} & \downarrow{\scriptstyle f} \\ & & Y \end{array}$$

be a commutative diagram of algebraic spaces over S where i and j are formally unramified and f is formally smooth. Then the canonical exact sequence

$$0 \to \mathcal{C}_{Z/Y} \to \mathcal{C}_{Z/X} \to i^*\Omega_{X/Y} \to 0$$

of Lemma lemma-two-unramified-morphisms is exact and locally split.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Denote $Z \to Z'$ the universal first order thickening of Z over X. Denote $Z \to Z''$ the universal first order thickening of Z over Y. By Lemma lemma-universally-unramified-differentials-sequence here is a canonical morphism $Z' \to Z''$ so that we have a commutative diagram

$$\begin{array}{ccccc} Z & \longrightarrow & Z' & \xrightarrow{a} & X \\ & \searrow^{i'} & \downarrow{\scriptstyle k} & & \downarrow{\scriptstyle f} \\ & & Z'' & \xrightarrow{b} & Y \end{array}$$

The sequence above is identified with the sequence

$$\mathcal{C}_{Z/Z''} \to \mathcal{C}_{Z/Z'} \to (i')^*\Omega_{Z'/Z''} \to 0$$

via our definitions concerning conormal sheaves of formally unramified morphisms. Let $U'' \to Z''$ be an étale morphism with U'' affine. Denote $U \to Z$ and $U' \to Z'$ the corresponding affine schemes étale over Z and Z'. As f is formally smooth there exists a morphism $h : U'' \to X$ which agrees with i on U and such that $f \circ h$ equals $b|_{U''}$. Since Z' is the universal first order thickening we obtain a unique morphism $g : U'' \to Z'$ such that $g = a \circ h$. The universal property of Z'' implies that $k \circ g$ is the inclusion map $U'' \to Z''$. Hence g is a left inverse to k. Picture

$$\begin{array}{ccc} U & \longrightarrow & Z' \\ \downarrow & \overset{g}{\nearrow} & \downarrow{\scriptstyle k} \\ U'' & \longrightarrow & Z'' \end{array}$$

Thus g induces a map $\mathcal{C}_{Z/Z'}|_U \to \mathcal{C}_{Z/Z''}|_U$ which is a left inverse to the map $\mathcal{C}_{Z/Z''} \to \mathcal{C}_{Z/Z'}$ over U. $\square$